MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 10559205 (INDICATOR, COMMAND, DISPLAY)

1. SCOPE

1.1 This specification covers one type of electronic assembly known as Circuit Card Assembly: 10559205, (Indicator, Command, Display) (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel: Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F10559205	Circuit Card Assembly-Indicator, Command, Display

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PACKAGING DATA SHEET

10559205 Packaging of Circuit Card Assembly: 10559205

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. - The circuit card assembly shall be manufactured in accordance with Drawing F10559205 and drawings pertaining thereto.

3.2 General specification. - The circuit card assembly shall meet the following requirements, where applicable, of MIL-F-13926:

(a) Order of precedence
(b) Dimensions and tolerances
(c) Part identification and marking

3.2.1 Electronic workmanship. - The circuit card assembly shall meet the workmanship requirements of Drawing F10559205 and drawings pertaining thereto.

3.3 Performance. - Unless otherwise specified, the circuit card assembly shall meet the performance requirements of this specification at standard ambient temperatures between 60 degrees Fahrenheit (^{o}F) and 90°F.

3.3.1 Loads, power and signals. - The circuit card assembly shall perform as specified herein when the loads, power, and signals of Table I are applied as specified.

TABLE I

ITEM	CONDITION	CHARACTERISTIC	CONNECTIONS
1.	Loads:		
1.1	Resistor and bias	620 ohms ±5%, 1/2W and ±15 ±0.5 Vdc	Connected in series between P1-13 and P1-J with negative return to P1-J

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TABLE I (continued)

ITEM	CONDITION	CHARACTERISTIC	CONNECTIONS
1.2	Resistor and bias	200 ohms ±5%, 2W and +15 ±0.5 Vdc	Connected in series between P1-P and P1-J with negative return to P1-J
2.	Power sources:		
2.1	5 Vdc	Tolerance: ±0.2 Vdc Ripple: 25 mV p-p maximum	Connected between P1-2(+) and P1-J(-)
2.2	15 Vdc	Tolerance: ±0.5 Vdc Ripple: 25 mV p-p maximum	Connected as specified in Items 1.1 and 1.2
2.3	220 Vdc	Tolerance: ±10 Vdc Ripple: 100 mV p-p maximum	Connected between P1-A(+) and P1-J(-)
3.	Signal sources:		
3.1	Digital type-A	Logical one: 4.0 ±1 Vdc Logical zero: 0.2 ±0.2 Vdc	Applied as specified in Tables II and III

TABLE II

NIXIE TUBE	TERMINAL	INPUT									
V1	P1-K	1	0	NA	NA	NA	NA	NA	NA	NA	NA
	V1 Readout No.	0	5	NA	NA	NA	NA	NA	NA	NA	NA
V2	P1-11	1	0	1	0	1	0	1	0	1	0
	P1-10	1	1	0	0	1	1	0	0	1	1
	P1-9	1	1	1	1	0	0	0	0	1	1

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TABLE II (continued)

NIXIE TUBE	TERMINAL	INPUT									
V2	P1-8	1	1	1	1	1	1	1	1	0	0
	V2 Readout No.	0	1	2	3	4	5	6	7	8	9
V3	P1-7	1	0	1	0	1	0	1	0	1	0
	P1-6	1	1	0	0	1	1	0	0	1	1
	P1-F	1	1	1	1	0	0	0	0	1	1
	P1-E	1	1	1	1	1	1	1	1	0	0
	V3 Readout No.	0	1	2	3	4	5	6	7	8	9
V4	P1-D	1	0	1	0	1	0	1	0	1	0
	P1-C	1	1	0	0	1	1	0	0	1	1
	P1-B	1	1	1	1	0	0	0	0	1	1
	P1-1	1	1	1	1	1	1	1	1	0	0
	V4 Readout No.	0	1	2	3	4	5	6	7	8	9
V5	P1-L	0	1	0	1	0	1	0	1	0	1
	P1-M	0	0	1	1	0	0	1	1	0	0
	P1-12	0	0	0	0	1	1	1	1	0	0
	P1-N	0	0	0	0	0	0	0	0	1	1
	V5 Readout No.	0	1	2	3	4	5	6	7	8	0

Legend: 1 - Logical one, type-A
0 - Logical zero, type-A
NA - Not applicable
All terminals are referenced to P1-J

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TABLE III

PERFORMANCE CONDITION	INPUT TERMINALS		VOLTAGE AT OUTPUT TERMINALS	
	P1-R	P1-14	P1-P	P1-13
1	0	0	1.2 ±0.7V*	1.2 ±0.7V*
2	0	1	1.2 ±0.7V*	15.0 ±0.8V
3	1	NA	15.0 ±0.8V	15.0 ±0.8V

* 1.2 ±0.9V at the temperature extremes specified in 3.3.2.2.

Legend: See Table II.

3.3.1.1 Readout selection logic.- The readout numbers for nixie tubes V1, V2, V3, V4, and V5 shall be as specified in Table II when the digital type-A signal of Table I is applied as indicated in Table II.

3.3.1.2 Indicator logic.- The voltages at the output terminals shall be as specified in Table III when the digital type-A signal of Table I is applied as indicated in Table III.

3.3.2 Environmental.

3.3.2.1 Storage temperature.- The circuit card assembly shall meet the requirements of 3.3.1 at ambient temperature (60°F to 90°F) after exposure and thermal stabilization at -80°F and+160°F.

3.3.2.2 Operating temperatures.- The circuit card assembly shall meet the requirements of 3.3.1 while exposed to and thermally stabilized at -40°F and +125°F, subsequent to which it shall meet the above requirements at ambient temperature (60°F to 90°F).

3.3.2.3 Vibration.- The circuit card assembly shall show no physical failure and shall meet the requirements of 3.3.1 after being exposed to the following vibratory conditions:

(a) Amplitude: 0.5 inch maximum
(b) Acceleration: 4 g maximum

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(c)	Crossover frequency:	12.5 Hertz (Hz)
(d)	Sweep time:	5 to 500 to 5 Hz in 15 minutes. Two sweeps in each of three mutually perpendicular axes

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. - The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval. - The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three circuit card assemblies. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. - When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing. - When the contractor is responsible for conducting first article approval tests, the sample shall be in inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

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4.3 Inspection provisions.

4.3.1 Submission of product. - Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies. - All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection. - Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables IV, V, and VI herein. Examination and test for packaging and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables IV, V, and VI shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.3.2.3 Functional tests. - The requirements and tests in Table IV shall be inspected on a 100% basis. All examinations and tests shall be conducted at the standard ambient temperature (60° to 90°F).

TABLE IV

(100 percent inspection)

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURES
101.	Readout selection logic	3.3.1.1	4.6.2
102.	Indicator logic	3.3.1.2	4.6.3

4.3.3 Acceptance and rejection. - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

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4.4 Special sampling.

4.4.1 General. - One circuit card assembly shall be selected at random by a Government representative as a control sample from each 100 produced. The control sample shall meet the requirements and tests in Table V.

TABLE V

	CONTROL TEST	REQUIREMENT	TEST PROCEDURES
301.	Fabrication	3.1	Applicable drawings - Visual
302.	General specification	3.2	MIL-F-13926 - Visual

4.4.2 Environmental. - Three circuit card assemblies shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall meet the requirements and tests in Table VI.

TABLE VI

	CONTROL TEST	REQUIREMENT	TEST PROCEDURES
303.	Storage temperature	3.3.2.1	4.6.4.1
304.	Operating temperature	3.3.2.2	4.6.4.1
305.	Vibration	3.3.2.3	4.6.4.2

4.4.3 Failure of sample. - Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted item has been approved (see 4.3.3).

4.5 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment. - Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

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4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with applicable technical data listed in the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.3 Test equipment.- In conjunction with 4.5.2, the standard test equipment listed in Table VII shall be utilized in performing the applicable test of 4.6.

TABLE VII

	NOMENCLATURE	DESCRIPTION
1.	D-C Power Supply	Plus 5 ±0.2V, Ripple: 25 mV p-p maximum
2.	D-C Power Supply	Plus 220 ±10V, Ripple: 100 mV p-p maximum
3.	D-C Power Supply	Plus 15 ±0.5V, Ripple: 25 mV p-p maximum
4.	D-C Voltmeter	Range of 0 to 30V, Accuracy of ±1%
5.	Signal Source	Capable of providing a d-c level of amplitude specified in item 3.1, Table I.

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4.6 Test methods and procedures.

4.6.1 Test conditions.- Unless otherwise specified, the test conditions shall be in accordance with the "Test Facilities" requirements of MIL-F-13926.

4.6.1.1 Initial conditions.- Connect the loads and power to the circuit card assembly as specified in Table I.

4.6.2 Readout selection logic.-

(a) Apply the initial conditions of 4.6.1.1.
(b) Apply the digital type-A signal of Table I of the logic level indicated, in turn, to the group of terminals associated with each nixie tube specified in Table II and observe the readout number on the nixie tube.

The nixie tube readout numbers shall be as specified in 3.3.1.1.

4.6.3 Indicator logic.

(a) Apply the initial conditions of 4.6.1.1.
(b) Apply the digital type-A signal of Table I of the logic level indicated, in turn, to the group of terminals indicated in Table III for each performance condition.
(c) Measure the voltages at the output terminals specified in Table III for each performance condition.

The output voltage shall be as specified in 3.3.1.2.

4.6.4 Environmental tests.- The test equipment shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 except that the circuit card assembly may be mounted on an inside surface of the test chamber for the temperature test and the tolerance requirements for vibration frequency measuring instruments shall not apply.

4.6.4.1 Temperature test.-

(a) Place the circuit card assembly in the test chamber and reduce the chamber temperature to -80°F (see note below).

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(b) Soak the circuit card assembly at this temperature for at least 5 hours with the unit not operating.
(c) Raise the chamber temperature to -40°F.
(d) Following thermal stabilization at -40°F, place the circuit card assembly in operation and perform the tests specified in 4.6.2 and 4.6.3.
(e) Raise the chamber temperature to +160°F.
(f) Soak the circuit card assembly at +160°F for at least 5 hours with the assembly not operating.
(g) Reduce the chamber temperature to +125°F.
(h) Following thermal stabilization at +125°F, place the circuit card assembly in operation and perform the tests specified in 4.6.2 and 4.6.3.
(i) Reduce the chamber temperature to ambient.
(j) Following stabilization at ambient temperature, place the circuit card assembly in operation and perform the test specified in 4.6.2 and 4.6.3.

The circuit card assembly shall meet the requirements of 3.3.2.1 and 3.3.2.2.

NOTE: The rate of temperature change shall be less than 7°F per minute during temperature cycling.

4.6.4.2 Vibration.- With the circuit card assembly firmly mounted to the vibration table, perform the following:

(a) Test in accordance with Method 514-I of MIL-STD-810 for category (f) equipment, except that the acceleration shall not exceed ±4 g and the maximum excursion (double amplitude) shall be 0.5 inch.
(b) Sweep the frequency from 5 Hz to 500 Hz and back to 5 Hz in a period of 15 minutes. Perform two sweeps in each of the three mutually perpendicular axes.
(c) Following vibration cycling, examine the circuit card assembly for structural failures, loose hardware, wire breakage and abrasion.
(d) Place the circuit card assembly in operation and perform the tests specified in 4.6.2 and 4.6.3.

The circuit card assembly shall meet the requirements of 3.3.2.3.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Packaging Data Sheet 10559205. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- This circuit card assembly is a plug-in module of the Command Display Indicator which energizes the range and reset indicators and displays range information on nixie tubes when supplied with appropriate combinations of digital input signals.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Applicable packaging data sheet number (see 5.1)
(c) Selection of applicable levels of preservation, packaging and packing
(d) Applicable stock number
(e) Requirement for First Article Submission.

Custodian:

Army - MU

Preparing Activity:

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Project No. 1240-A140

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